ASEQUOIA FLUX

Academic Research Overview and Theoretical Foundations

Continuous-relaxation heuristics for the Euclidean Traveling Salesperson Problem

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RESEARCH STATUS | EXPLORATORY

Not peer-reviewed. No proof of P = NP. Global optimality, polynomial-time solvability, and comparative performance have not been established.

Purpose of This Directory

This directory brings together two exploratory manuscripts that investigate continuous-relaxation heuristics for the Euclidean Traveling Salesperson Problem (TSP). The manuscripts share a broad motivation—using continuous edge weights as an intermediate search representation—but differ in their mathematical mechanisms and reference implementations.

The work is a research proposal, not an established solver. Neither manuscript proves that the TSP can be solved in polynomial time, resolves the P versus NP question, or establishes a global-optimality guarantee. The purpose of this overview is to describe the proposals accurately, distinguish implemented mechanisms from hypotheses, and identify the tests needed to evaluate whether the methods offer practical value.

Objective and Scope

The Euclidean TSP asks for a shortest closed tour through a set of cities in which each city is visited exactly once. Enumerating all possible tours gives a factorial-size search space, but this does not describe every TSP algorithm: exact methods and heuristics exploit additional mathematical and computational structure. TSP is NP-hard, and a continuous representation alone does not remove that difficulty.

The project explores whether continuous edge-activation variables, constraint feedback, regularization, and a separate discrete reconstruction stage can provide a useful heuristic for small or moderate Euclidean instances. Its contribution at this stage is a pair of explicit, testable formulations—not a proof that continuous dynamics overcome NP-hardness.

Featured Manuscripts

1. A Physics-Inspired Continuous-Relaxation Heuristic for the Euclidean Traveling Salesperson Problem

This manuscript uses physical language as a source of design intuition. Its reference formulation combines four mechanisms:

1. **Cost continuation.** An affine interpolation of normalized edge costs is used as a continuation device. It is not a Gaussian convolution and does not establish measure preservation, Wasserstein invariance, or preservation of a global minimizer.
2. **Degree correction and active suction.** Primal-dual feedback and a degree-residual penalty encourage each city to have total incident edge weight near two. This is a heuristic constraint mechanism, not a guarantee of exact feasibility at every integration step.
3. **Geometric crossing penalty.** Candidate pairs of straight-line edges that properly intersect are penalized when both receive weight. This is a geometric penalty, not a Coulomb inverse-square force, and it does not guarantee a crossing-free final tour.
4. **Double-well crystallization.** A quartic penalty, proportional to $x^2(1-x)^2$, encourages edge weights toward zero or one. It does not by itself guarantee integrality or Hamiltonicity.

The reference workflow separates continuous optimization from discrete tour construction. The relaxed weights bias a multi-start greedy construction, after which 2-opt local search improves the candidate tours using the original Euclidean distances. The reconstructed output is a valid Hamiltonian cycle when the procedure completes normally, but it is not guaranteed to be optimal.

Key limitation: preprocessing all pairs of candidate edges to detect geometric crossings can require $O(n^4)$ work in the worst case. The implementation should therefore not be described as a demonstrated scalable polynomial-time TSP solver.

2. A Degree-Constrained Algebraic Relaxation Heuristic for the Euclidean Traveling Salesperson Problem

This manuscript expresses the proposal using a symmetric continuous edge-weight matrix and an explicit degree-constrained feasible set. The main mechanisms are:

1. **Degree-constrained fractional 2-factor relaxation.** The variables satisfy symmetry, zero diagonal, bounds $0 \le x_{ij} \le 1$, and weighted degree constraints $\Sigma_j x_{ij} = 2$. This is not the Birkhoff–von Neumann polytope, whose row and column sums equal one, and it is not the full TSP polytope.
2. **Spectral connectivity regularization.** The second-smallest eigenvalue λ_2 of the weighted graph Laplacian is used to encourage connectivity. With a positive softening parameter ε, the proposed reciprocal penalty remains finite at disconnection; it is a soft penalty rather than an infinite barrier.

3. **Quartic binary penalty.** The term $\Sigma\, x_{ij}^2(1 - x_{ij})^2$ encourages edge weights toward zero or one but does not guarantee binary values.
4. **Projected updates and parameter continuation.** Updates are projected back toward the degree-constrained feasible set while regularization weights change over iterations. The reference implementation uses numerical optimization for the projection, making it a small-instance research prototype rather than a large-scale implementation.

The relaxation can represent fractional states and disconnected 2-factors. Spectral connectivity is not equivalent to Hamiltonicity, and a connected weighted graph is not necessarily a Hamiltonian cycle. A separate tour-reconstruction procedure is therefore included; tour validity and tour quality must be evaluated independently of the relaxation matrix.

Key limitation: the non-convex objective, the spectral term's possible non-differentiability when eigenvalues are repeated, and the numerical projection step prevent the current formulation from supporting a general convergence or polynomial-time guarantee.

Shared Mathematical Status

The following distinctions should be kept explicit when describing, citing, or extending this work.

What the formulations specify

- Continuous edge weights and their intended interpretation.
- Degree-related constraints or feedback, together with regularization terms.
- A parameter schedule intended to change the relative strength of the terms during optimization.
- A separate conversion from continuous weights to a discrete tour.
- Testable implementation choices and known limitations.

What has not been established

- **Global optimality:** no proof shows that either method returns an optimal TSP tour.
- **Global convergence:** no complete Lyapunov analysis or convergence theorem is provided for the time-varying, non-convex dynamics.
- **Homotopy guarantees:** no proof rules out singular Hessians, branch changes, or bifurcations along the parameter schedule.
- **Integrality:** the double-well penalties encourage binary values but do not prove that all variables converge to zero or one while satisfying the other constraints.
- **Hamiltonicity:** degree constraints and spectral connectivity regularization do not, by themselves, enforce a single valid Hamiltonian cycle.
- **Polynomial-time solvability:** a polynomial cost per iteration would not suffice. One would also need a justified bound on the number of iterations and a guarantee about the solution quality or correctness. Such a bound has not been established here.
- **P versus NP:** neither manuscript resolves or claims to resolve P = NP.

Preliminary Status and Evidence

The current work should be treated as exploratory. The supplied drafts do not contain a systematic benchmark study that establishes solution quality, runtime scaling, or an advantage over established baselines. Qualitative behaviour observed in preliminary runs, if reported, is not sufficient evidence for a complexity claim or for general algorithmic effectiveness.

A reproducible evaluation should include:

- **Baselines:** nearest neighbour, nearest neighbour plus 2-opt, and a strong TSP heuristic such as LKH; use an exact solver on small cases where an optimal value is available.
- **Instances:** controlled random Euclidean instances and suitable TSPLIB instances, with instance identifiers and random seeds recorded.
- **Metrics:** tour validity, tour length, relative gap to an optimum or best-known value, wall-clock runtime, and run-to-run variability.
- **Ablations:** remove each mechanism in turn to measure whether cost continuation, degree feedback, crossing penalties, spectral regularization, or binary penalties improve results beyond the reconstruction baseline.
- **Protocol:** tune hyperparameters on separate instances, fix stopping criteria before testing, and publish code and raw results.

Complexity should be reported for the complete pipeline, not only for an individual gradient or matrix operation. In the physics-inspired reference implementation, geometric crossing-pair preprocessing has $O(n^4)$ worst-case cost. In the algebraic formulation, dense eigenvalue calculations can cost $O(n^3)$ per evaluation, and the numerical projection's runtime depends on the optimization routine and instance. Neither fact alone gives a bound on the number of iterations required to reach a desired tour quality.

Invitation for Peer Review

Feedback from theoretical computer science, applied mathematics, and operations research is welcome, especially on the following questions:

- Are the objectives, gradients, projections, and stopping criteria implemented consistently with their mathematical definitions?
- Can small counterexamples expose failure modes in the degree feedback, spectral regularization, or continuation schedule?
- Does the continuous stage improve tour quality beyond multi-start greedy construction and 2-opt alone?
- How does the method compare with established baselines when runtime, hyperparameter tuning, and reconstruction cost are all included?
- Which mechanisms should be retained, revised, or removed based on ablation results?

Negative results and counterexamples are valuable. The aim is to determine whether these formulations provide measurable practical value—or whether their proposed mechanisms fail to improve on simpler approaches.

Suggested Repository Contents

File	Purpose
Improved_Physics_Inspired_TSP_Paper.pdf	Physics-inspired formulation, limitations, and reference workflow.
Improved_Algebraic_TSP_Paper.pdf	Degree-constrained relaxation, spectral regularization, and continuation formulation.
physics_tsp_reference.py	Reference implementation for the physics-inspired heuristic.
algebraic_relaxation_tsp.py	Reference implementation for the algebraic relaxation heuristic.
Academic_Research_Theoretical_Foundations_Overview.pdf	This research overview and statement of status.

File names in an actual release should be updated to match the files distributed in that release. Do not list a script or manuscript as included unless it is present and reproducible in the repository.

Conclusion

The Asequoia Flux research project investigates two continuous-relaxation heuristics for Euclidean TSP. One emphasizes degree feedback, geometric crossing penalties, and staged binary regularization; the other emphasizes a degree-constrained fractional 2-factor relaxation, spectral connectivity regularization, and projected updates. Their common idea is to use a continuous matrix as an intermediate representation before reconstructing and improving a discrete tour.

At present, these are exploratory methods. They have no demonstrated global-optimality guarantee, no established polynomial-time bound for reaching a target solution quality, and no systematic benchmark evidence of superiority over standard TSP heuristics. Their scientific value should be judged by transparent mathematical definitions, reproducible implementations, careful baselines, ablation studies, and independent scrutiny.

Status: Exploratory research. Not peer-reviewed. Not a proof of P = NP. Empirical validation remains necessary.

Selected References

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- Fiedler, M. “Algebraic Connectivity of Graphs.” *Czechoslovak Mathematical Journal*, 23(2), 298–305, 1973.
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- Tero, A., et al. “Rules for Biologically Inspired Adaptive Network Design.” *Science*, 327(5964), 439–442, 2010.